Collision lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test momentum conservation with measured before-and-after velocities.

SUCCESS CRITERIA - I can...

1. Record masses, signs and velocities consistently.
2. Compare total momentum before and after.
3. Discuss external impulse and measurement variation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “momentum” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

momentum: mass times velocity, including direction

external: acting from outside the chosen system

01 Warm-up: bring back an earlier idea

A **3 kg** object slows from **7** to **3 m/s** in **1 s**.
1. **Find impulse** and average net force.
2. Repeat the force calculation for the same velocity change over **2 s**.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total} = m_1v_1 + m_2v_2$; $\Delta p_{total} = p_{total,\,f} - p_{total,\,i}$. **Compare just before**/after collision. Use approved carts; **record actual trials separately** from practice data.

Worked reasoning: Supplied trials give total momenta **6.0** before and **5.8 kg·m/s** after. The change is **-0.2 kg·m/s**. **Compare this** with measurement uncertainty before attributing it to a new physical effect.

Your turn: Draw a before-and-after momentum comparison and **name one possible external interaction**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Supplied case: cart A is **0.50 kg** at **+0.40 m/s**; B is **0.50 kg** at rest. After sticking, both move at **+0.20 m/s**. **Find total momentum before** and after.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Cart	Before (m/s)	After (m/s)
A: 0.50 kg	+0.40	+0.20
B: 0.50 kg	0	+0.20

Marzano check - circle: ① ② ③

04 Apply the physics

In a lab collision, a **0.50 kg** cart moving at **+0.60 m/s** sticks to a stationary **0.75 kg** cart. The measured common final velocity is **+0.23 m/s**. **Complete the table**, find total momentum before and after, and **calculate the percent deviation relative** to the initial momentum.

Sentence starter: I found … [unit]. This value means … in this situation.

Cart	Mass (kg)	v before (m/s)	v after (m/s)
A	0.50	+0.60	+0.23
B	0.75	0	+0.23

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

1. Design a table for three trials with masses and signed velocities before/after.
2. **Name one possible external impulse** and one measurement limitation.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Launch at **40 m/s** from level ground at **30°** or **60°**. Use g=**10** and no drag. **Compare ranges** and flight times. **Label separate horizontal** and vertical motion.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

07 Exit check

A nonzero initial momentum is **0.40 kg·m/s** and final is **0.38**. **Find percent deviation** and **state why this** is not automatically a **2%** uncertainty.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Initial total is **2.0±0.1** and final total **1.8±0.1 kg·m/s**. **Find worst-case bounds** for the change. If contact lasts **0.2 s**, bound average external force.

Sentence starter: I found … [unit]. This value means … in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③